MRIDUL VERMA R

AI & Fullstack Engineer

Gmail — LinkedIn — GitHub — LeetCode

Summary

AI and Full Stack Developer with hands-on experience building end-to-end applications using MERN Stack and deploying ML models to production. Strong foundation in DSA, REST APIs, and CI/CD. Interested in building scalable systems with AI-driven features such as chatbot assistants, image classification, and smart recommendation workflows.

Technical Skills

Programming: Java, Python, JavaScript (ES6)
Frontend: React.js, HTML, CSS, Tailwind CSS
Backend: Node.js, Express.js, REST APIs, JWT Auth
Database: MongoDB, Mongoose
AI/ML: CNN, Transfer Learning, Model Training, Inference Pipelines
Tools & DevOps: Git/GitHub, Docker (Basics), AWS (EC2, S3), CI/CD (CodePipeline/CodeDeploy)

Projects

Farmer AI Assistant Sep 2025 – Nov 2025

- Built CNN-based crop disease identification system for leaf images with high accuracy.
- Implemented image preprocessing (resize, normalization, augmentation) to improve model generalization.
- Developed a voice-supported multilingual chatbot supporting Tamil, Hindi, and English.
- Integrated live market price API and weather-based crop recommendations for decision support.
- Designed backend APIs to store user queries, predictions, and history for better tracking.
- Improved usability for farmers by keeping the workflow simple: capture image → diagnosis → guidance.

Cravely – Full Stack Food Ordering Platform Nov 2025 – Dec 2025

- Architected a responsive MERN application with cart, orders, and real-time order tracking.
- Developed RESTful APIs using Express.js and implemented JWT authentication for secure access.
- Added role-based access for users, restaurants, and admin-side management features.
- Implemented WebSocket-based live order updates to track preparation and delivery status.
- Integrated payment workflow and handled order confirmations with proper status updates.
- Optimized MongoDB queries and indexing to improve performance by 30%.

DevOps: Jenkins Pipeline for EKS Cluster Provisioning January 2026

- Created Jenkins CI/CD pipeline automating Terraform-based AWS EKS cluster provisioning.
- Integrated Trivy security scanning to detect Terraform IaC vulnerabilities before deployment.
- Implemented multi-stage pipeline: Plan → Security Scan → Apply → Destroy with approval gates.
- Automated end-to-end infrastructure lifecycle for create, update, and destroy operations.
- Used parameterized builds for environment selection (dev/test) and reusable pipeline execution.
- Managed GitHub repository for version control, collaboration, and tracking infrastructure changes.

Experience

AI/ML Developer Intern Oct 2025 – Nov 2025
ShadowFox Erode, India

- Completed end-to-end ML workflows: data collection, preprocessing, model development, evaluation, and production deployment
- Developed production-ready ML systems for real-world agricultural applications
- Gained hands-on experience transitioning ML models from experimentation to production

Education

B.E. Computer Science & Engineering Aug 2023 – May 2027
Bannari Amman Institute of Technology
CGPA: 8.74

HSLC Aug 2022 – Mar 2023
Carmel Matric Higher Secondary School
Grade: 96.8

Achievements & Certifications

- 300+ LeetCode Problems Solved showcasing strong problem-solving and optimization skills
- NPTEL: Responsible and Safe AI Systems
- Oracle AI Foundations Associate (2025)
- Cisco Introduction to Cybersecurity